Matthew Hutchinson

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Mechanical Engineer focused on physical AI: combining robot learning, autonomy, and embedded systems with hands-on hardware development and testing.

EDUCATION

UCLA | Master of Science in Mechanical Engineering (Robotics & AI) September 2025 – August 2026

Reinforcement Learning, Deep Learning, Computer Vision, Autonomous Aerospace Systems, Soft Robotics, Bionic Systems, Natural Language Processing, Compliant Mechanisms, Data Science

UCLA | Bachelor of Science in Mechanical Engineering September 2022 – June 2024

Robotics (Kinematics, Dynamics & Control), Rapid Prototyping, Mechanism Design, Data Structures & Algorithms, C++, Python

EXPERIENCE

UCLA Robotics & Mechanisms Laboratory (RoMeLa) June 2026 – Present

Research Intern

- Built a Python/ROS 2 control and diagnostics stack for an 18-actuator hexapod, integrating dual CAN buses, calibration, telemetry logging, fault handling, and synchronized 100 Hz motion
- Developed and tested coordinated leg motion, walking gaits, and artistic choreography using motor feedback and bounded trajectories
- Led robot's electrical redesign by characterizing power requirements and specifying the fused distribution, wiring, contactor/E-stop architecture, connectors, and component BOM for the museum platform and next-generation robot

UCLA DevX September 2022 – June 2024

Autonomous Rover Developer

- Owned autonomy and behavior stack for a competition rover: built line-following logic, lost-line recovery, and intersection handling within a ROS 2-based perception + control system (Raspberry Pi + Arduino)
- Implemented 30 Hz wheel-speed feedback using a PID structure with PD gains (Ki=0); tuned using step-response tests under varying loads and battery conditions
- Sized drive motors for a 15° ramp with payload; achieved some full-course runs under 5 minutes, but camera-dependent performance remained inconsistent
- Integrated electronics subsystem: spec'd battery pack, wired buck converter and 10A fuse; mitigated camera latency by reducing resolution and implementing adaptive velocity scaling

UCLA Engineering MakerSpace April 2024 – July 2024

Digital Fabrication Artist

- Mentored 30+ students through design thinking and hands-on prototyping (laser cutters, 3D printers, CNC tools); led workshops on Illustrator and Fusion 360 and authored quick-reference guides that reduced equipment downtime

NASA Langley Research Center August 2022

Intern, AAM Safety Team

- Investigated safety and airspace integration challenges for Advanced Air Mobility and eVTOL aircraft; proposed a conceptual detect-and-avoid (DAA) architecture using neural networks and multi-sensor fusion, delivering a safety assurance framework to NASA engineers

SKILLS

Languages & Tools: Python, MATLAB, C++, ROS 2, PyTorch, OpenCV, GCP Compute Engine, Git, Arduino, SolidWorks

Robotics & Systems: PID control, embedded systems (Raspberry Pi, Arduino), hardware integration, digital fabrication (laser cutting, 3D printing)